Leon Riccius | Curriculum Vitae

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leon-riccius

Education

Doctor of Philosophy — **Delft University of Technology, NL**
Civil Engineering — *07/2021 – present*

Thesis: Efficient Bayesian Inference of Parametric Partial Differential Equations
Supervisors: Dr. Iuri Rocha & Dr. Frans P. van der Meer

Master of Science — **Technical University Munich, GER**
Mechanical Engineering — *03/2018 – 05/2021*

Thesis: Machine Learning Augmented Turbulence Modelling for the Reynolds Stress Closure Problem
Supervisor: Prof. P.S. Koutsourelakis

Bachelor of Science — **Technical University Munich, GER**
Mechanical Engineering — *10/2013 – 02/2018*

Thesis: Diffeomorphic Matching on Thrombus and Outer Surfaces of Abdominal Aortic Aneurysms
Supervisor: Prof. M.W. Gee

Semesters abroad:
Keio University, Tokyo, JP — *03/2019 – 09/2019*
Ecole Polytechnique de Montréal, Montréal, CA — *01/2016 – 05/2016*

Research

My research develops efficient sampling algorithms for complex probability distributions. These continuous-time sampling algorithms are based on piecewise-deterministic dynamics, leveraging score function evaluations and gradient-based event rates — mathematical machinery shared by modern generative models (diffusion processes, flow matching).

Selected publications

[1] Leon Riccius, Atul Agrawal, and Phaedon-Stelios Koutsourelakis. Physics-Informed Tensor Basis Neural Network for Turbulence Closure Modeling. In *37th Conference on Neural Information Processing Systems, NeurIPS*, 2023.

[2] Leon Riccius, Iuri Rocha, Joris Bierkens, Hanne Kekkonen, and Frans van der Meer. Integration of Active Learning and MCMC Sampling for Efficient Bayesian Calibration of Mechanical Properties. *European Journal of Mechanics A/Solids*, 2026.

[3] Leon Riccius, Iuri Rocha, Joris Bierkens, Hanne Kekkonen, and Frans van der Meer. Piecewise Deterministic Markov Processes for Bayesian Inference of PDE Coefficients. 2026. under review.

I presented my work at 7 international conferences, including SIAM AN, ECCOMAS, and UNCECOMP.

Machine Learning Experience

Modeling:	Gaussian process, Neural Networks, Variational Autoencoders (VAE), Conditional VAEs
Software & Tools:	Python (pytorch, jax), `C++`, bash, git
Communication:	Data visualization, academic writing, public speaking, supervision

Github Repositories

SLIMM-Lab/pdmp: Library for approximating complex probability distributions via piecewise deterministic Markov process sampling (pytorch based ML, jax based differentiable FEM models)
SLIMM-Lab/mcmc-with-surrogates: Bayesian inference of FEM model parameters where expensive likelihoods are approximated with Gaussian process models (`C++` based)

Awards

Travel Award **Montréal, CA**
Society for Industrial and Applied Mathematics (SIAM) *2025*

Best Poster Award **Papendal, NL**
27th Engineering Mechanics Symposium (114 submissions) *2024*

Teaching

TU Delft, Faculty for Civil Engineering and Geoscience **Delft, NL**
Designing interactive textbook[1], teaching assistant[2], project supervisor[3] *07/2021 – present*

- Modeling, Uncertainty, and Data Analysis for Engineers[1,2,3]
- Data Science and Artificial Intelligence for Engineers[1,2,3]
- Statistical Learning for Engineers[2]

Supervision:
Daan Smolders, Master Thesis, in cooperation with Airbus 2024
Surrogate modelling for uncertainty propagation by targeting regions of high variance and probability

Group projects:

- Bayesian optimization of microstructure topologies of porous media (2023)
- Composite microstructure generation with conditional variational autoencoders (2022)
- Comparing Bayesian machine learning techniques for predicting finite element solutions (2021)

Languages

German:	Native proficiency	
English:	Full professional proficiency	*TOEFL iBT Score: 111*
French:	Limited working proficiency	*Abitur: B2/C1*
Dutch:	Elementary proficiency	
Japanese:	Elementary proficiency	